

Red Blood Cell Positional History, Sphericity, and Network Architecture Influence Linging and Partitioning

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ABSTRACT The transport and distribution of red blood cells (RBCs) through microvascular networks is central to diverse applications in drug delivery, tissue engineering, and the study of pathological mechanisms. Recent findings highlight two critical phenomena: (i) a history-dependent partitioning effect, wherein the RBC partitioning at each bifurcation is influenced by the positional and configurational state inherited from previous bifurcations and (ii) RBC lingering at the bifurcation apex, which strongly affects downstream flow distribution. However, the interplay between network geometry, RBC sphericity (in spherocytosis disease), and these emergent behaviors remains insufficiently explored. Here, we perform two-dimensional Lattice Boltzmann simulations coupled with an immersed boundary model to systematically investigate RBC dynamics in both asymmetric and fully symmetric network designs. Our results show that RBC positional history grows progressively with each bifurcation, leading to increasing deviations from classical partitioning laws regardless of the network's geometric complexity. We further demonstrate that even spherical RBCs can experience substantial lingering due to cumulative apex interactions, challenging the notion that only elongated or pathologically deformed cells are prone to these effects. These findings clarify how RBC deformability and prior apex encounters affect microcirculatory flow.

SIGNIFICANCE How red blood cells navigate the microvasculature is critical for oxygen delivery. We used simulations to show that an RBC's path is not random but is influenced by its past interactions at vessel bifurcations. We found this memory effect accumulates, causing RBCs to linger longer and deviate more from classical flow laws at each successive bifurcation. This phenomenon is robust, persisting even in perfectly symmetric networks, and affects pathological (spherical) cells as well as healthy ones. These findings provide a new biophysical framework for how cell history and deformability govern blood perfusion, which is crucial for understanding microcirculatory disorders.

INTRODUCTION

The blood circulatory system, with its hierarchical structure of vessels ranging from large arteries to capillaries, presents many aspects which challenge our current understanding. Among these, the behavior of RBCs within this complex network is highly important. RBCs are not simple carriers of oxygen; their behavior, flexibility, and aggregate formation significantly regulate vascular blood flow, and therefore the distribution of oxygen throughout the body (1). It has been suggested from recent studies that RBCs possess a history-dependent behavior—that is, their past experiences within the vascular network, including shear exposure and passing through narrow capillaries or interacting with the endothelium, might influence their future behavior and distribution (2–4).

Throughout this paper we use the term “positional memory” or “RBC history effect” to denote the phenomenon whereby an RBC arriving at a bifurcation carries information about its prior lateral position and deformation state from upstream bifurcation encounters. Our objective is to investigate how the positional memory of RBCs influences their microcirculation and especially whether it leads to increased lingering of RBCs in specific regions of the network (apex of the bifurcations), which could be affecting tissue oxygenation and the efficiency of metabolic waste removal. Furthermore, the partitioning of RBCs—specifically how they are distributed and rerouted at vascular bifurcations—could also be influenced by this memory,

with significant implications for the understanding of microcirculatory disorders, and which can also help the design of artificial blood substitutes or microfluidic devices for biomedical applications.

The partitioning law or the Zweifach-Fung effect, named after the researchers who first described it, addresses the non-uniform distribution of blood components at vascular bifurcations. The core message of this effect is that, at a bifurcation, the fraction of RBCs entering the high-flow daughter branch is larger than the fraction of total blood flow entering that branch—that is, $N^* > Q^*$ when $Q^* > 0.5$ —meaning the high-flow branch is enriched with RBCs, beyond what a simple flow-rate split would predict (5). The diameters of daughter vessels relative to the mother vessel modulate the magnitude of this enrichment, in addition to other factors like the hematocrit and the flow stagnation point in the bifurcations.

Actually, there are two different types of partitioning: (i) The classical partitioning (the Zweifach-Fung effect, as described above) (ii) The reversed partitioning, which means (4) that the RBC split at a bifurcation is *less* extreme than the flow split—specifically when $N^* < Q^*$ for $Q^* > 0.5$ (see Table 2). Note that reversed partitioning does *not* necessarily mean that the majority of RBCs enter the lower-flow branch; it simply means that the fraction of RBCs directed to the high-flow branch falls below that branch's fractional flow. In extreme cases, the majority of RBCs can indeed enter the low-flow branch, but milder degrees of reversed partitioning—where most RBCs still go to the high-flow branch but proportionally less than expected—are equally possible (4). Conditions favoring reversed partitioning include a significantly larger vessel diameter in the lower-flow branch, or specific microcirculatory conditions such as upstream positional bias of the RBCs.

Lingering of RBCs refers to the prolonged retention or slow transit of red blood cells within the bifurcation apex of the microvascular network (see Fig. 1d). This behavior contrasts with the expected swift passage of RBCs, driven by the pressure and flow dynamics typical of healthy circulation. Lingering can result from a variety of factors, including vascular architecture, RBC deformability, local hematocrit variations, and the presence of biochemical signals within the microenvironment (6, 7).

There are many factors influencing RBC dynamics, like vascular geometry and network complexity, where the design of the microvascular network, including vessel diameter, bifurcation angles, and network density, can significantly impact RBC transit times (8). Areas with complex branching patterns and narrow bifurcation apices may promote lingering by creating regions of reduced flow velocity near bifurcation tips (7). At the low Reynolds numbers, characteristic of microcirculation, vessel tortuosity itself plays a negligible role (9); it is primarily the bifurcation geometry and RBC deformability that govern lingering. For example, in the case of RBC deformability, their very flexible nature allows them to navigate the narrow capillaries and bifurcations. However, changes in RBC deformability, due to disease or aging, can hinder their ability to move efficiently through the microvasculature, leading to increased retention in certain areas. The variations in local hematocrit levels, also influenced by the partitioning behavior at bifurcations and the dynamic adjustment of blood flow, can lead to heterogeneous RBC distribution and contribute to lingering, particularly in regions with reduced flow (6, 7, 10). The microvascular endothelium secretes numerous signals that could influence the behavior of RBCs by modulating the degree of vasodilation or vasoconstriction of a vessel and the adhesiveness of RBCs to the walls. These biochemical cues can change the characteristics of local flow and transit times of RBCs (11).

The concept that upstream bifurcations induce cross-sectional skewness, thereby altering downstream RBC partitioning, has been well established in recent years. Impressive 3D direct numerical simulations in realistic, asymmetric microvascular networks (10, 12) have demonstrated how upstream events and RBC jamming create skewed hematocrit profiles. These findings are complemented by *in vitro* network studies (13) and detailed quantifications of RBC lingering at individual bifurcations (6). However, in such complex networks, it is inherently difficult to decouple the intrinsic positional history of the cells from local geometric biases, such as asymmetric flow splits and vessel tapering. Furthermore, these prior large-scale studies have focused almost exclusively on healthy, highly deformable discocytes. In the present study, we address these gaps by systematically quantifying compounding deviations in partitioning and lingering level-by-level. By introducing a perfectly symmetric network as a control, we definitively isolate intrinsic RBC positional history from geometric bias. Additionally, we extend this framework to spherical RBCs (characteristic of spherocytosis), revealing a distinct, counterintuitive lingering mechanism driven by multi-cell geometric blockage rather than the transient membrane hooking observed in healthy cells.

This paper will consider in detail the mechanisms of the positional-memory effect of RBCs and how it influences cell lingering and partitioning in complex vascular networks. We consider two complex networks, one artificial and symmetric, and one corresponding to the cat mesentery. Several important features are reported here regarding both partition and lingering. We study the case of biconcave shapes as well as more inflated cells, the extreme case correspond to quasi-spherical shape, such as

that encountered in spherocytosis diseases. The major outcomes of our study are as follows:

- We find a significant deviation from the empirical partition law(14) for healthy RBCs, confirming previous works (10, 12, 15). We also confirm the occurrence of inverse partition(4). Similar conclusions are found here for spehrocytes, a previously not studied case.
- We find ample lingering for healthy RBCs and show that the cell positional memory upstream singnificantly affects the result. We are not aware of any previous study showing the role of bifurcations cascade leading ultimately to large lingering amplitude.
- Importantly, this finding helps resolve a previously reported discrepancy between simulations and in vivo experiments. In Rashidi et al.(6), 3D numerical simulations were compared with in vivo measurements, and significantly larger lingering values were observed experimentally than in the simulations. However, those 3D simulations considered only a single bifurcation. In contrast, our study incorporates multiple successive bifurcation levels and demonstrates that lingering effects and deviations from classical partitioning laws increase cumulatively after each bifurcation. This cumulative mechanism naturally explains why in vivo observations exhibit much larger lingering values: although measurements are performed locally, the cells have already traversed many upstream bifurcations outside the observation field. Importantly, our 2D simulations show quantitative agreement with the in vivo experimental trends.
- Surpisingly, spherocytes show quantitatively similar values of the lingering effect. It is usually thought that the flexibility of healthy cells is mainly responsible for lingering. Indeed, when a cell impacts the apex of a bifurcation, its flexibility allows it to spread out and conform to the shape of the apex. This delays the cell at the apex before it begins its journey towards one of the two daughter branches. However, spherocytes, being close to a spherical shape, possess very low flexibility, but despite this, the rate of lingering is quite comparable to that observed in healthy cells. We will present an explanation to this finding based on the ample contribution of cell crowding at the apex for spherocytes.
- We find that the precise structure of the network plays a relatively minor role. Indeed, we did not see any noticeable difference between the two studied network (an artificial symmetric network, and the one corresponding to mesentery). We will provide some basic ideas underlying this behavior.

MATHEMATICAL MODEL AND SIMULATION METHOD

The simulation method is based on the Lattice Boltzmann Method (LBM) using a 2D geometry. The LBM is combined with the immersed boundary method for fluid–structure coupling. The RBC is modeled as a two-dimensional elastic vesicle: its membrane is characterized by a bending rigidity κ and a constraint of local inextensibility. Membrane forces (derived from the Helfrich energy functional (bending and tension contributions)) are spread onto the fluid grid via the immersed boundary method, so that the fluid solver sees the membrane as a distributed body force. The internal cytoplasm and external plasma have different viscosities, captured by the viscosity contrast $\lambda = \mu_{\text{in}}/\mu_{\text{ex}} = 6$, representative of physiological conditions. Full mathematical details of the membrane model and the LBM–IBM coupling are provided in Supplementary Section I. The LBM has been described and validated in prior works (11, 16–19). The validation of the LBM involved also a comparison with the boundary integral method (20). Several studies using 3D models, which include the membrane cytoskeleton, have demonstrated that the 2D model effectively replicates several shapes and flow behaviors of erythrocytes (21–24). Furthermore, we will see here that our 2D results capture not only qualitatively but also quantitatively the experimental observation regarding lingering(6). To further demonstrate the translational validity of 2D simulations, we refer to one of our previous work (4). In that study using a 2D simulation we had predicted for the first time the occurrence of an inverse partitioning. That work incited an experimental study (4) on RBCs using the same geometry as in the simulations. The obtained experimental results remarkably agreed with the 2D prediction.

All typical values of physical parameters used in this study and the values of dimensionless parameters are given in Table 1. Note that the plasma viscosity is taken equal to 0.001 Pa.s, instead of 0.0015. Pa.s. Actually it is the viscosity contrast which is

essential.

A remark on the parameter choices in Table 1 is warranted regarding the *Reynolds number*. Re is defined as $Re = \rho R_0^2 \dot{\gamma} / \mu_{ex}$, where R_0 is the characteristic RBC radius and $\dot{\gamma}$ is the characteristic shear rate. The physiological Re in microcirculation is of order 10^{-3} or below. The LBM, however, inherently incorporates finite inertial effects and becomes numerically unstable at arbitrarily small Re . Previous validations of the present numerical framework have shown that $Re = 0.1$ is sufficiently small for inertial contributions to be negligible and for Stokes-regime physics to be faithfully recovered (11, 17). We therefore set $Re = 0.1$ throughout the present simulations.

In all simulations, flow was driven by a fixed body force (equivalent to a constant pressure gradient) applied to all fluid nodes. No-slip boundary conditions at vessel walls were enforced via the standard halfway bounce-back scheme. Periodic boundary conditions were applied along the flow direction, linking the outlet vessels directly back to the network inlet. RBCs were initialized with their equilibrium (stress-free) shape and introduced at the network at a fixed hematocrit of 20%, with

Table 1: Physical and dimensionless parameters used in the simulations.

Physical Parameters	
Cell Radius R_0 (μm)	3
Membrane Bending Modulus κ (J)	3×10^{-19}
Fluid Density ρ (kg m^{-3})	1×10^3
Plasma Viscosity μ_{ex} (Pa s)	1×10^{-3}
Dimensionless Parameters	
Reynolds Number Re	0.1
Capillary Number Ca	25
Viscosity Contrast λ	6
Reduced Area τ	0.45 to 0.99

random initial transverse positions. Each simulation was run until a statistically stationary state was reached (typically after 1000 times the relaxation time of the RBC), and all reported statistics (lingering Péclet numbers, flux fractions, hematocrit values, and PDF profiles) were collected exclusively during this steady-state phase. Blood flow rates in individual vessel segments were computed by integrating the velocity field over a mid-vessel cross-section perpendicular to the mean flow direction, and local hematocrits were measured by counting the number of RBCs crossing a reference cross-section over a fixed time window.

RBCs partitioning

We will consider two complex networks (see later). Our first focus will be to deal with the RBCs partitioning at bifurcations. As stated in the introduction, classical partitioning (the Zweifach-Fung effect) refers to the situation where a *disproportionately* large fraction of RBCs enters the branch with the highest flow rate ($N^* > Q^*$ when $Q^* > 0.5$). More recently, reversed partitioning has been reported, in which the RBC split is less extreme than the flow split ($N^* < Q^*$ when $Q^* > 0.5$) (4). Let us consider a given bifurcation (Fig. 1b) where two daughter branches are denoted as ‘1’ and ‘2’. The fractional blood flow Q^* and the fractional RBCs flow N^* are defined as:

$$N^* = \frac{N_{\text{daughter1}}}{N_{\text{daughter1}} + N_{\text{daughter2}}}$$

$$Q^* = \frac{Q_{\text{daughter1}}}{Q_{\text{daughter1}} + Q_{\text{daughter2}}}, \quad (1)$$

where $N_{\text{daughter1}}$ and $N_{\text{daughter2}}$ are the number of RBCs that cross a chosen section in the first and the second daughter branches respectively, and $Q_{\text{daughter1}}$ and $Q_{\text{daughter2}}$ are the blood flow rates in the first and the second daughter branches respectively. Table 2 summarizes the conditions for the classical and the reverse partitioning.

Table 2: Identifying the partitioning type based on Q^* and N^* (10).

	$Q^* > N^*$	$Q^* < N^*$
$Q^* > 0.5$	Reverse partitioning	Classical partitioning
$Q^* < 0.5$	Classical partitioning	Reverse partitioning

An empirical law was developed by Pries *et al.* (14) regarding RBCs partitioning in a bifurcation, relating Q^* and N^* (this

empirical law was derived from direct experimental observation of RBC partitioning in microvascular bifurcations, not from an assumed partitioning type):

$$N^* = \begin{cases} 0, & \text{if } Q^* < X_0, \\ 1, & \text{if } Q^* > 1 - X_0, \\ \frac{1}{1 + e^{-\left[A + B \logit\left(\frac{Q^* - X_0}{1 - 2X_0}\right)\right]}}, & \text{otherwise.} \end{cases} \quad (2)$$

where $\logit(x) = \ln\left(\frac{x}{1-x}\right)$, A , B and X_0 are related to the geometry of bifurcation and hematocrit H :

$$A = \frac{-6.96}{D_M} \ln\left(\frac{D_{MD}}{D_{SD}}\right), B = 1 + 6.98 \left(\frac{1-H_M}{D_M}\right), \quad X_0 = 0.4/D_M, \quad (3)$$

The subscripts MD refer to the main daughter (that which recruits more RBCs), SD (for secondary) to the other one and M to the mother branch. H is the hematocrit, D the vessel diameter (or width in 2D). Here, the hematocrit refers to the discharge hematocrit, which is related to the tube hematocrit. We also note that the difference between the tube and discharge hematocrits was negligible.

Note that in the symmetric network the vessel diameters are constant across all levels, so that A and X_0 remain fixed. The hematocrit of the mother vessel H_M changes with each bifurcation level as the RBC distribution evolves downstream, which modifies B and hence the slope of the sigmoid. In the mesentery network, both diameters and hematocrits vary with the bifurcation level. In all comparisons between simulation and Eq. (2), we used the actual mother-vessel diameter and discharge hematocrit measured at each individual bifurcation, so that the Pries et al. (14) baseline is correctly adapted to the local conditions at every level.

Our aim will be to compare the results resulting from simulation to relation (2). Let us denote by:

$$\Delta N^* = N^*_{MD} - N^*_{SD} \quad (4)$$

the difference in RBCs number crossing the main and secondary branch. This quantity will be computed numerically and will be denoted as ΔN^*_{SIM} and will be compared to the one obtained from the empirical law (2), denoted as ΔN^*_{EMP} , and to quantify the deviation from the Zweifach-Fung prediction, we calculated the deviation d_{SE} from the theoretical prediction using Eq.(2) as the relative difference between this prediction and our simulation (15):

$$d_{SE} = |\Delta N^*_{SIM} - \Delta N^*_{EMP}| \quad (5)$$

RBCs lingering

The calculation method for lingering, as described in previous research (25), is briefly summarized here for completeness. To quantify lingering, a relative residence time, Pe_L , is defined as the normalized time an RBC spends at the intersection between the mother branch and the two daughter branches (highlighted in yellow in Fig. 1.b):

$$Pe_L = \frac{t_{r,RBC}}{t_{ref}}, \quad (6)$$

where $t_{r,RBC}$ is the residence time of a specific RBC at the yellow intersection, and $t_{ref} = \frac{L}{V_M}$ is the reference time required for an RBC to travel the reference length L (from the exit of the ROI of the mother branch (blue area) to the apex and then to the entry of the daughter branches; Fig. 1.b)) with a velocity equal to the mean velocity at the exit of the mother branch (V_M).

This definition is similar to the lingering Péclet number defined by Rashidi et al. (15), except that it is a cell-based definition rather than a statistical estimation based on the probability density function of RBC velocity. This approach was also tested in our simulations, yielding similar trends (See Fig. S2 and Fig. S3 in the supplementary materials). As in (25), an RBC is considered to be lingering if $Pe_L \geq 2.0$.

We note that Pe_L is defined as a ratio of actual residence time to a reference time computed from the mean flow velocity in the mother vessel. This construction makes Pe_L robust to reductions in mean flow rate: if a cell simply travels more slowly because the bulk flow is slower (as occurs at deeper network levels where the flow splits repeatedly), both $t_{r,RBC}$ and t_{ref} scale similarly, and Pe_L remains of order unity. An increase in Pe_L above 2 therefore genuinely indicates that the cell resides at the apex for *longer than expected* for its local flow conditions, and is not merely an artifact of reduced flow velocity.

FIGURE 1

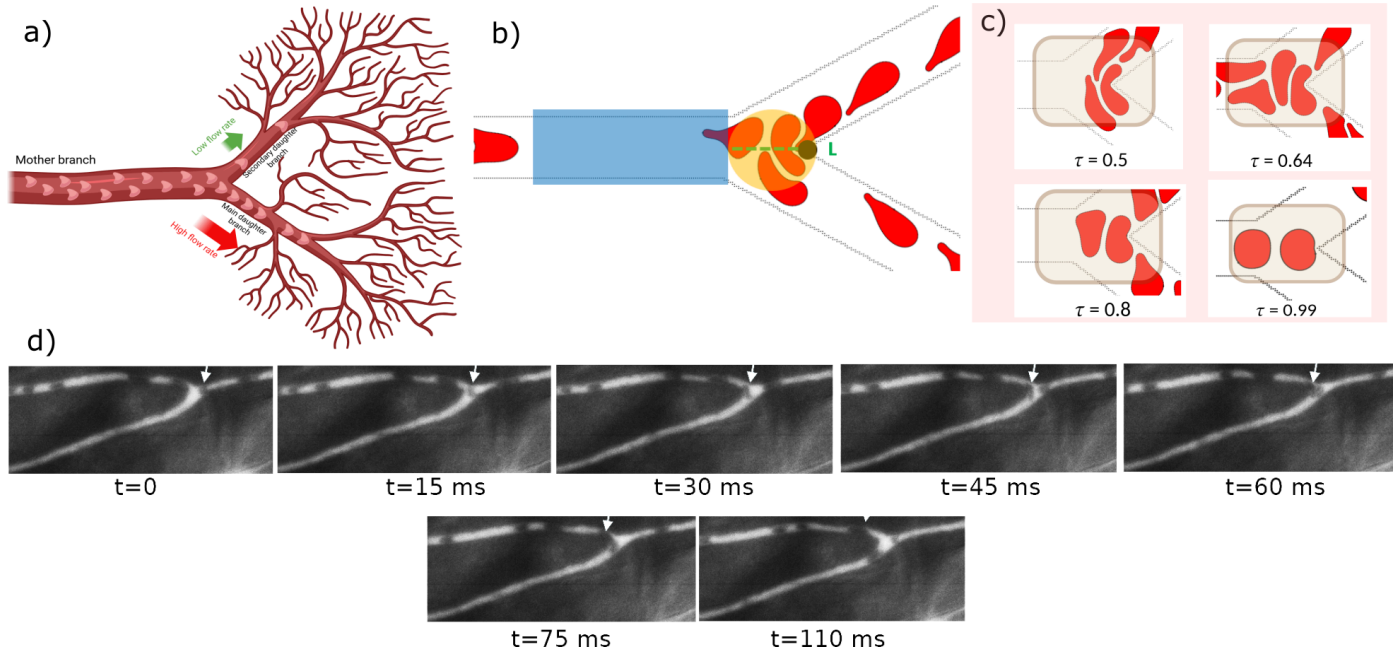


Figure 1. Overview of RBC partitioning and lingering in microvascular bifurcations. (Asymmetric partitioning) a) A schematic of a bifurcation showing a classical partitioning, the Mother branch (M), the Main Daughter branch (MD) as the vessel collecting the highest number of RBCs and the Secondary Daughter branch (SD) as the other vessel. b) Schematic of a bifurcation example: the blue area represents the mother branch, while the intersection of this area with the daughter branches is highlighted in yellow. The reference length L is defined as the distance from the exit of the blue area to the apex of the bifurcation. c) Linging snapshots near a given bifurcation for different reduced area. d) Linging in a real microvascular network (15).

Geometries

Here, we used two different complex geometries with multiple bifurcation levels (Fig. 2). In the first complex geometry extracted from a cat mesentery as depicted in Fig. 2.a and adopted from (12), there are three levels of bifurcations in series from an arteriole to capillaries and then the confluence of the capillaries into a venule. In total, this network comprises 7 bifurcations and 22 vessels. The radii of the smallest capillary and feeding artery are $7\mu\text{m}$ and $20\mu\text{m}$, respectively.

The second network in Fig. 2.b that we generated consists of four levels of bifurcations, having 15 bifurcations and 32 vessels in total. The width of all vessels is uniform, around $8.4\mu\text{m}$. Unlike the first network, which has asymmetric branching, the geometry of each bifurcation in the second network is symmetric. We note that the hematocrit is set to 20% and the capillary number to 25, while the viscosity contact for the RBCs is 6.

FIGURE 2

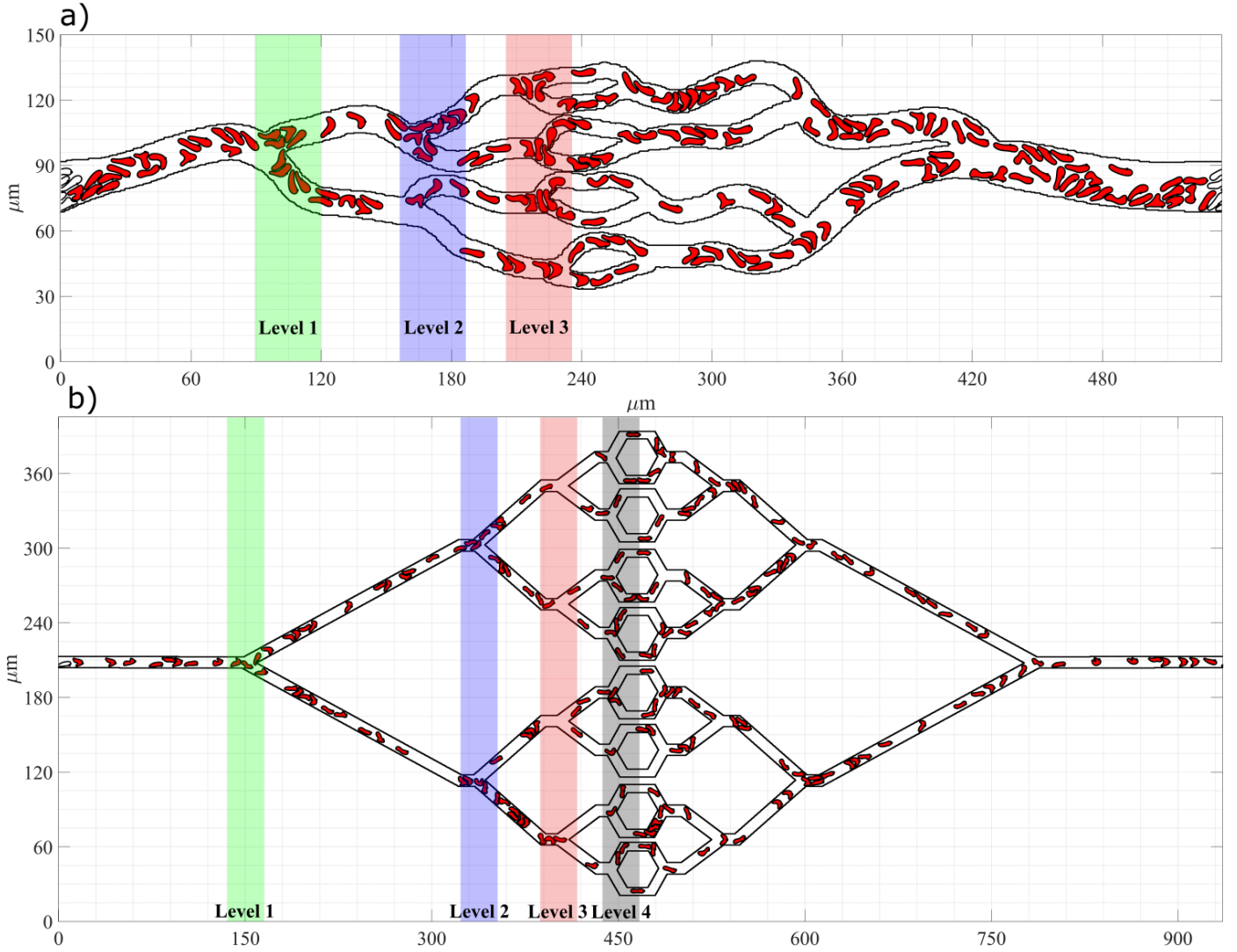


Figure 2. (a) Asymmetric network: Complex geometry extracted from cat mesentery with 3 bifurcation levels taken from (12). (b) Symmetric network: Complex symmetric geometry with 4 bifurcation levels.

RESULTS AND DISCUSSION

Lingering and Deviation from empirical law across different network levels for normal cells (discocytes) and spherocytes

We present two key metrics (the lingering Péclet number, Pe_L , and the deviation from the empirical law (2) d_{SE}) as functions of the bifurcation level in two different microvascular networks of Fig. 2.

Deviation from the empirical Law (d_{SE})

In Fig. 3 we provide the raw simulation results (symbols) obtained for different reduced area (as described in Table 1) and at different bifurcation levels of the two networks presented in Fig. 2. The blue line corresponds to the empirical law (2).

We examine now more precisely how the deviation from the empirical law evolve in successive bifurcations of the networks. We consider the two extreme cases of healthy cells and the case of spherocytes (Fig. 4), in order to see whether the shape has or not a significant impact. Figs. 4a (asymmetric network) and b (symmetric network) show the boxplots of d_{SE} , which quantifies the difference between the simulated RBC flux distribution and the empirical law (2) for discocytes. In both networks, the deviation d_{SE} grows with each additional bifurcation level. For the first network (Fig. 4(a)), it increases from around 0.05 (Level 1) to 0.1 (Level 2) and surpasses 0.20 at Level 3. In the second network (Fig. 4(b)), d_{SE} increases substantially from nearly 0 at Level 1 to around 0.6 at Level 4. This indicates that standard partitioning laws become less accurate further into the microvascular network, likely due to the compounded effects of RBC lingering, altered flow rates, and RBC deformability (3, 10).

These findings are qualitatively consistent with the experimental observations of Mantegazza et al. (13), who reported similar scatter and increasing deviations from Pries' law at successive bifurcation levels in an *in vitro* microchannel network (their Fig. 9a). The present simulations provide a mechanistic explanation for that observation: the progressive accumulation of positional bias in RBC distributions, driven by repeated apex interactions, systematically shifts the RBC flux fractions away from the flow-ratio distribution predicted by the empirical law.

We also note from Fig. 3 that a fraction of the simulation data points fall below the empirical curve (i.e. $N^* < Q^*$ for $Q^* > 0.5$), corresponding to reversed partitioning events. The proportion of such events grows with bifurcation level, consistent with the increasing positional asymmetry documented in Fig. 6. This means that the positional history of RBCs not only amplifies classical partitioning deviations but can also trigger conditions favoring reversed partitioning at downstream bifurcations.

The case of spherocytes is shown in Fig. 4 c,d (asymmetric and symmetric network, respectively). Here again we see a notable deviation from the empirical law, albeit slightly smaller than in the discocyte case. Below we will provide some qualitative explanation leading to an increasing deviation from the empirical law in successive bifurcations.

FIGURE 3

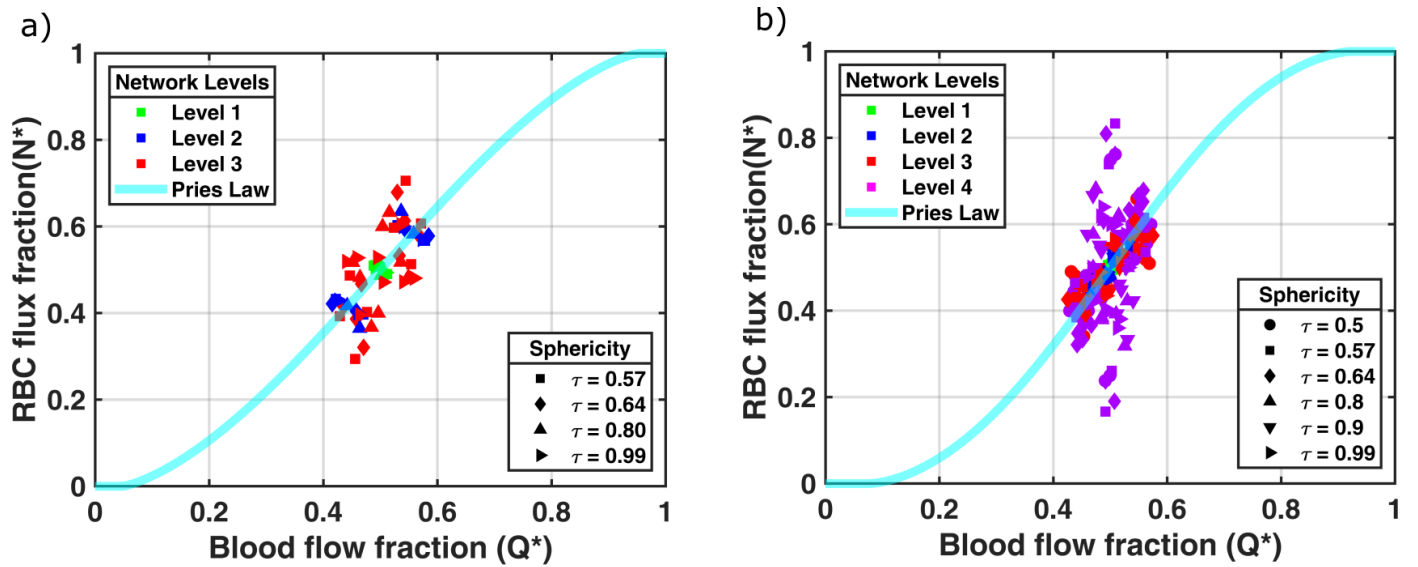


Figure 3. Blood flow fraction Q^* vs. RBC flux fraction N^* for various reduced areas across multiple bifurcation levels. Points above the empirical curve (blue line) correspond to classical partitioning ($N^* > Q^*$ for $Q^* > 0.5$), while points below correspond to reversed partitioning ($N^* < Q^*$ for $Q^* > 0.5$). (a) Results for the asymmetric network; (b) results for the symmetric network.

FIGURE 4

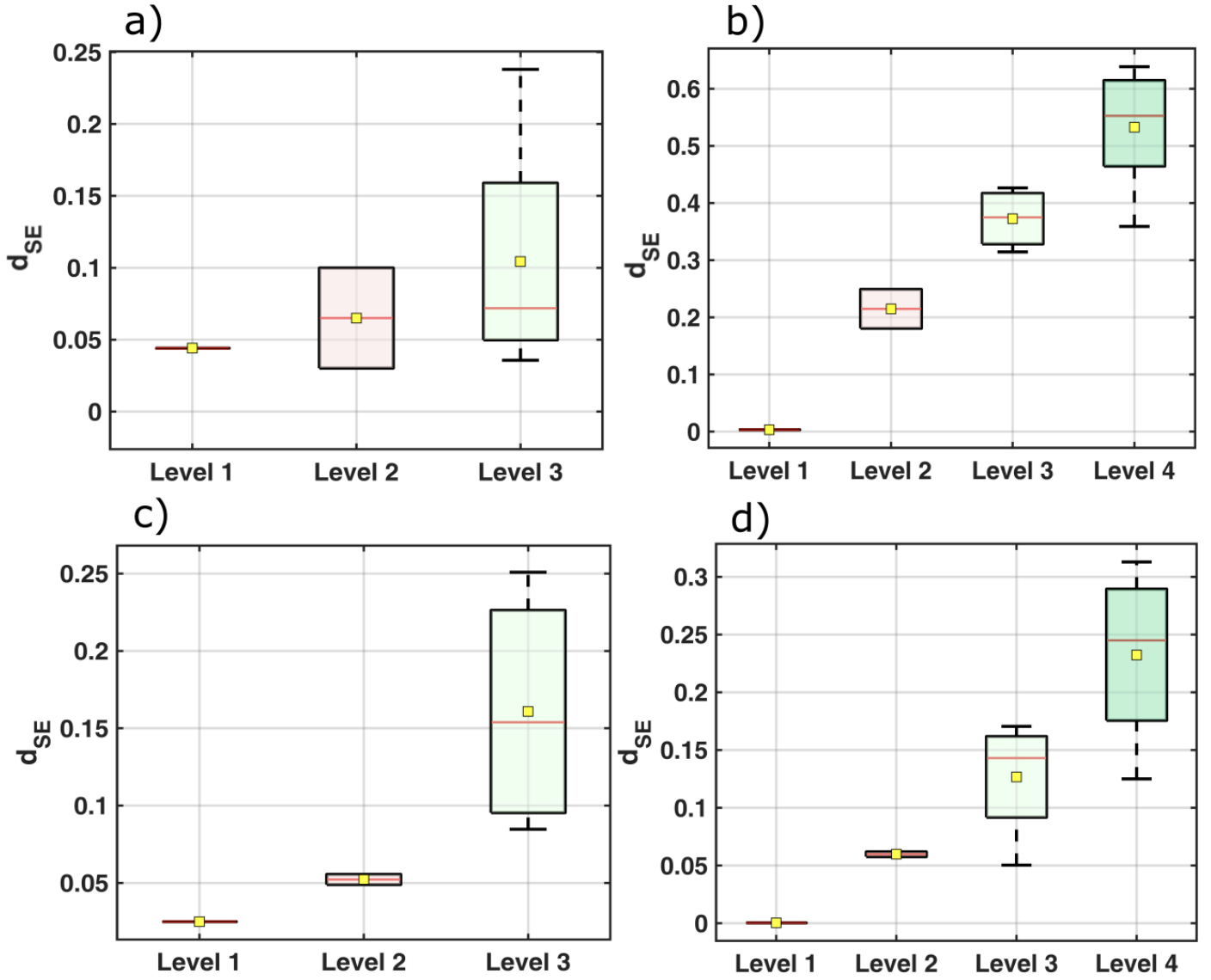


Figure 4. Boxplots of d_{SE} (deviation from the empirical law): (a) asymmetric network with three levels for normal cells ($\tau = 0.64$); (b) symmetric network with four levels for $\tau = 0.64$; (c) asymmetric network for spherical cells ($\tau = 0.99$); and (d) symmetric network with four levels for $\tau = 0.99$. Boxes indicate the interquartile range (IQR), red lines denote the median, and squares represent mean values at each level.

Lingering Péclet number (Pe_L)

In Fig. 5(a), the average lingering Péclet number Pe_L increases notably from Level 1 to Level 3. Specifically, we observe median values around 1.7 at Level 1, increasing to ~ 1.9 at Level 2, and exceeding 2.2 at Level 3. This trend indicates that red blood cells tend to spend increasingly more time in the apex region of each bifurcation as they progress deeper into the network. A similar pattern emerges in Fig. 5(b), where Pe_L rises from ≈ 1.8 at Level 1 to over 2.6 at Level 4 in the second, symmetric network. This systematic rise in lingering is consistent with the notion of an RBC positional-memory effect, wherein cells' past deformation and apex interactions at upstream bifurcations alter their dynamics in subsequent levels (6, 25). Interestingly, in Fig. 5(c,d) we observe both qualitatively and quantitatively the same trend for spherocytes. This is a priori unexpected, since it is often believed that the lingering is due to the large deformation of discocytes, which can significantly spread along the bifurcation apex and then takes some time to retract and detach from the apex. We will provide below some intuitive support for the reason why spherocytes can also experience a significant lingering.

We also want to highlight that the value of 'lingering', which is a primary focus of our investigation, is notably low in the 3D simulations presented by Rashidi et al.(6)(who used a single bifurcation), reaching a maximum of only around 0.6. In contrast, their experimental data showed values up to 2. This experimental maximum is highly consistent with the findings from our 2D

simulations (for the asymmetrical network): as stated above we observed values of approximately 2.2 using the new lingering method (detailed in the main text equation (6)) and up to 1.9 using the method described by Rashidi et al. (6) (Figures are in the supplementary material, and the equation is discussed in the Rashidi et al. paper(6), section “Lingering and partitioning”). This strongly suggests that our 2D simulations, which point the effect of successive bifurcations, capture the essential features.

FIGURE 5

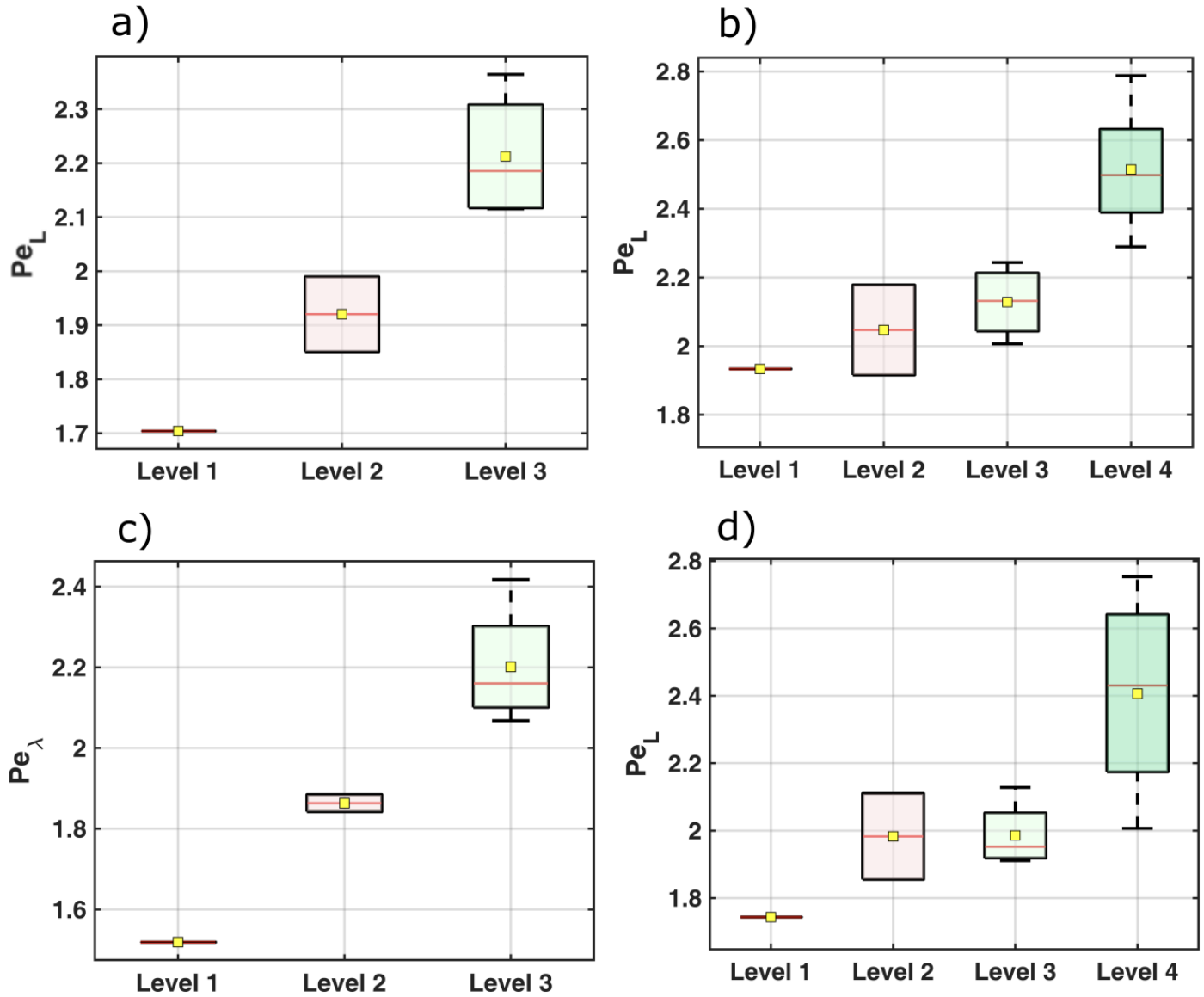


Figure 5. Boxplots of Pe_L (lingering Péclet number): (a) asymmetric network, normal cells ($\tau = 0.64$); (b) symmetric network ($\tau = 0.64$); (c) asymmetric network, spherical cells ($\tau = 0.99$); and (d) symmetric network ($\tau = 0.99$). Boxes indicate the interquartile range (IQR), red lines denote the median, and squares represent mean values at each level.

Symmetric Network Analysis and the Role of RBC history

In the previous section, we established that RBC lingering and deviations from the empirical law grow with each successive bifurcation. However, because the original mesenteric network in Fig. 2(a) includes both geometric asymmetry and vessel tapering (halving vessel widths after each bifurcation), it remains unclear whether these trends stem primarily from the network’s design or from an intrinsic history effect of the RBCs. To disentangle these factors, we constructed a purely symmetric network with constant vessel width (Fig. 2(b)). By comparing RBC behavior in these two extreme cases (an asymmetric, tapering network vs. a fully symmetric, constant-width network), we can determine whether RBC history alone is sufficient to produce the growing lingering and partitioning deviations observed at each successive bifurcation.

RBC Probability Density and Symmetry Index

Figs. 6(a)–(d) illustrate the RBC probability density function (PDF) measured in a transverse slice of the mother branch, just upstream of a bifurcation, for two different values of the reduced area τ (0.64 and 0.99) and at two different bifurcation levels (1st vs. 4th). As τ decreases (i.e., the RBC becomes more elongated), the PDF becomes increasingly asymmetric (see profile in the 4th level in Figs. 6(b)), suggesting that radial drift and apex interactions encountered in the previous bifurcations are actively biasing the RBC distribution. Even in this purely *symmetric* geometry, RBCs exhibit gradually increasing asymmetry as the level number grows, pointing to a positional history effect intrinsic to the RBCs rather than a mere byproduct of geometric asymmetry.

To quantify this effect, we define a spatial symmetry index,

$$Id_{sym} = 1 - \frac{MAE}{Data\ Range}, \quad (7)$$

where MAE is the mean absolute difference between the PDF values and their mirrored counterparts across the channel's centreline, and *Data Range* is the difference between the maximum and minimum PDF values observed across the channel width. Normalizing by the data range rather than by the absolute mean ensures that $\langle Id_{sym} \rangle$ is a scale-free quantity bounded between 0 and 1, regardless of the local hematocrit level: a value of 1 indicates perfect left–right symmetry (zero MAE), while values approaching 0 correspond to strongly asymmetric distributions. As shown in Fig. 6(e), $\langle Id_{sym} \rangle$ decreases markedly with each subsequent bifurcation level for all examined reduced areas. This indicates that the RBC distribution becomes less symmetric as cells progress through the network. Elongated cells (e.g., $\tau = 0.50$ and $\tau = 0.64$) exhibit stronger asymmetry shifts than their more spherical counterparts (e.g., $\tau = 0.99$) or near-spherical RBCs (e.g., $\tau = 0.80$).

Hematocrit Differences between daughter branches and Deviations at Symmetric Bifurcations

In Fig. 6(f), we track the absolute difference in hematocrit, d_{Ht} , between each pair of daughter vessels at bifurcations. Similar to the findings in Fig. 5, the inter-branch hematocrit difference increases with the bifurcation level, further confirming that the RBC distribution is progressively altered by upstream events. When daughter vessels carry unequal hematocrits, the more concentrated branch experiences increased cell–cell interactions and crowding near the next bifurcation apex. This crowding prolongs the residence time of individual cells (increased lingering) and simultaneously biases the flux fraction entering the next bifurcation away from the flow-based prediction of the empirical law (increased d_{SE}). Thus, hematocrit imbalances generated at one level self-consistently amplify both lingering and partitioning deviations at subsequent levels.

Global RBC Distribution and Plasma Skimming

The overall RBC distributions for $\tau = 0.64$ and $\tau = 0.99$ are shown in Fig. 6(g) and Fig. 6(h), respectively. In both cases, the majority of RBCs concentrate around the channel center, while near-wall regions show reduced hematocrit—a form of plasma skimming. In addition, these plots reveal that RBCs with a lower reduced area (more elongated shape) develop pronounced localized aggregations or “hotspots” at successive bifurcation levels, likely from the cumulative radial drift and apex interactions. As a result, even in a perfectly symmetric geometry with constant vessel widths, RBC distributions become increasingly imbalanced downstream.

FIGURE 6

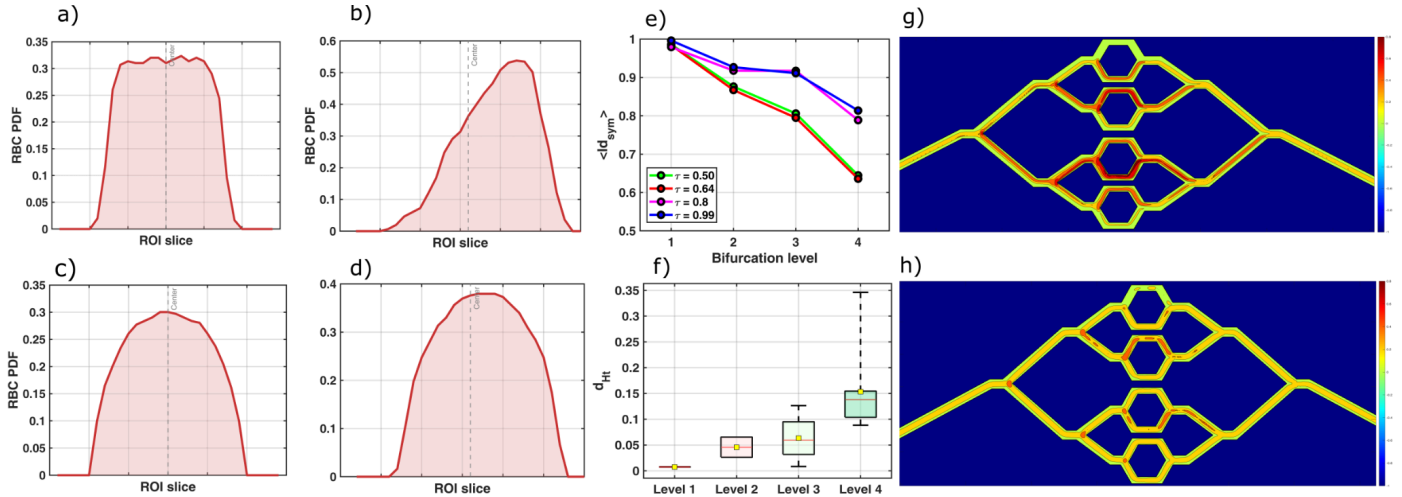


Figure 6. RBC distributions and hematocrit variations in the symmetric network. (a) Probability density function (PDF) of RBCs across the width of the mother branch (transverse channel position) in the *first* bifurcation level of the *symmetric network* for $\tau = 0.64$. (b) Same PDF measurement for the *fourth* level at $\tau = 0.64$. (c) PDF in the *first* level for $\tau = 0.99$. (d) PDF in the *fourth* level for $\tau = 0.99$. (e) Average symmetry index $\langle Id_{sym} \rangle$ of the RBC PDF versus bifurcation level for different reduced areas τ in the symmetric network. (f) Differences in hematocrit (d_{Ht}) between each pair of daughter vessels, also across the four bifurcation levels of the symmetric network. (g) Global distribution of RBCs ($\tau = 0.64$) in the entire symmetric network, with color indicating RBC concentration. (h) Same global distribution for $\tau = 0.99$.

Qualitative discussion of the results

Let us now discuss qualitatively the results reported in Fig. 6. For that purpose, let us consider a given RBC configuration in the symmetric network, as shown in Fig. 7. A cell arriving at the apex of the bifurcation of level 2, for example, will stay after the bifurcation in the vicinity of the two walls forming the bifurcation apex (see configurations between level 2 and level 3). The lift force tends to push the cells towards the center of the channel, but that requires a significantly longer distance (26–30). The same scenario occurs in successive bifurcations. This explains why the cell distribution is asymmetric, as shown in Fig. 6(b,d). This strong asymmetry also explains the deviation from the empirical law, which in a perfectly symmetric network should predict equal hematocrits in two daughter branches. In other words, the cell retains a positional “memory” of its encounter with the first bifurcation apex: its transverse position in the downstream mother vessel is biased toward the wall, and this bias is carried forward to subsequent bifurcations.

On the other hand, in the upper branch between level 2 and level 3 in Fig. 7, due to the aforementioned asymmetry of RBC configuration, most cells will go to the lower branch after the level 3 bifurcation. The same reasoning applies to the lower branch between level 2 and level 3, implying that at the next bifurcation most cells go to the upper branch. This means that extreme branches at the top or bottom of the network receive progressively fewer cells. This explains the results shown in Fig. 6(g,h): plasma skimming.

Finally, let us discuss qualitatively the occurrence of lingering. Consider first the case of discocytes. As shown in Fig. 8, due to the aforementioned asymmetry of the RBC configuration, most cells select the lower branch. However, when the lower branch becomes overly concentrated with RBCs, the local viscous resistance increases, reducing the effective pressure drop available to drive cells into that branch. Successive cells (e.g. the green cell in Fig. 8) are consequently redirected toward the bifurcation apex. Owing to their deformability, such cells spread along the apex geometry before retracting—a process that requires additional time and constitutes lingering. Once they detach, they typically select the upper (less concentrated) branch. This event is repeated regularly and leads to a certain proportion of cells which undergo a delay, or lingering, at bifurcations. Note that spherocyte deformation is small compared to discocytes, but still we observe in Fig. 5 that their lingering is quite significant. The explanations provided earlier clearly show that the positional history of RBCs plays an essential role: what happens, for example, at the level 1 bifurcation in Fig. 7, dictates the uneven partitioning, asymmetric distribution, plasma skimming, and so on.

FIGURE 7

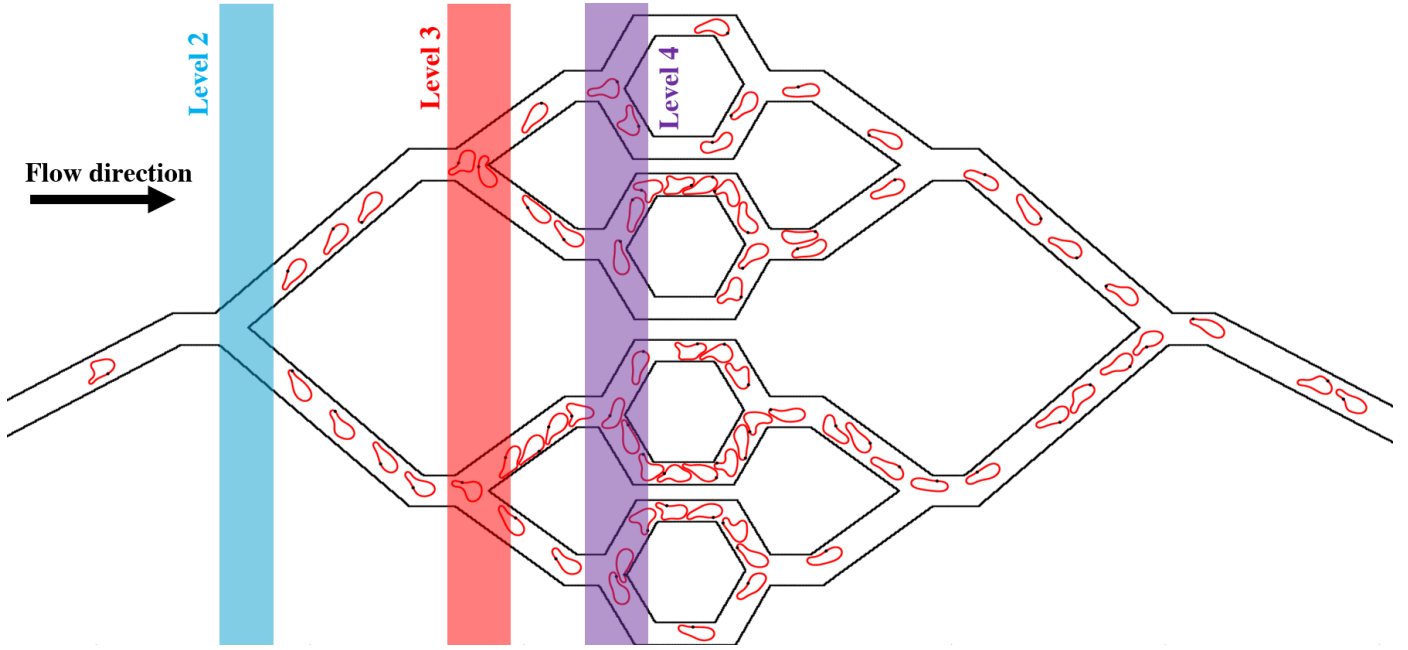


Figure 7. Visualization of the red blood cell distribution at the symmetric network for healthy cells across levels 2, 3, and 4.

FIGURE 8

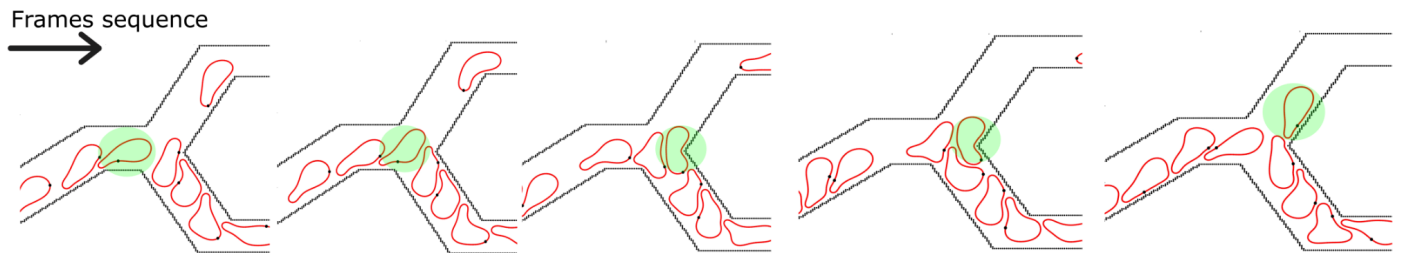


Figure 8. Frames snapshot of an RBC near the inner wall (highlighted in green) lingering at the apex of the bifurcation before flowing into the opposite daughter branch, away from the daughter branch closest to the inner wall (Level 4 of the symmetric network for $\tau = 0.64$).

Lingering Trends in Asymmetric and Symmetric Networks for Various Reduced Area

In this section we provide some quantitative results about lingering, and more precisely how the proximity of a cell to the apex favors lingering. We will also analyze the proportion of cells that undergo lingering; we recall that a cell is considered to be lingering if its Péclet number is equal to or higher than 2. This study will consider various reduced areas. Let D_L be the smallest distance from the mass center to the apex of a bifurcation.

- **Impact of Apex Proximity on Lingering.** Fig. 9 compares lingering behavior for the symmetric network. It shows bin-scatter plots of the lingering Péclet number Pe_L (vertical axis) as a function of D_L , for discocytes and spherocytes (other reduced areas and the asymmetric network are shown in supplemental Fig. S4). In each bin, the color scale indicates the fraction of RBCs that exhibit a particular (D_L, Pe_L) pairing. Cells closer to the bifurcation apex ($D_L \lesssim 2 \mu\text{m}$) have a higher probability of lingering ($Pe_L \geq 2$). This trend is slightly more pronounced for discocytes, which are more elongated and can more easily physically hook onto the apex boundary (i.e. a portion of the elongated cell membrane temporarily wraps around the bifurcation tip, creating a transient mechanical contact that delays detachment—hereafter referred to as *physical hooking*). Spherical RBCs ($\tau = 0.99$) also show elevated Pe_L values when D_L is very small. As discussed in the section *Lingering Behavior of Spherical RBCs* below, their main lingering mechanism is blockage events rather than physical hooking.

FIGURE 9

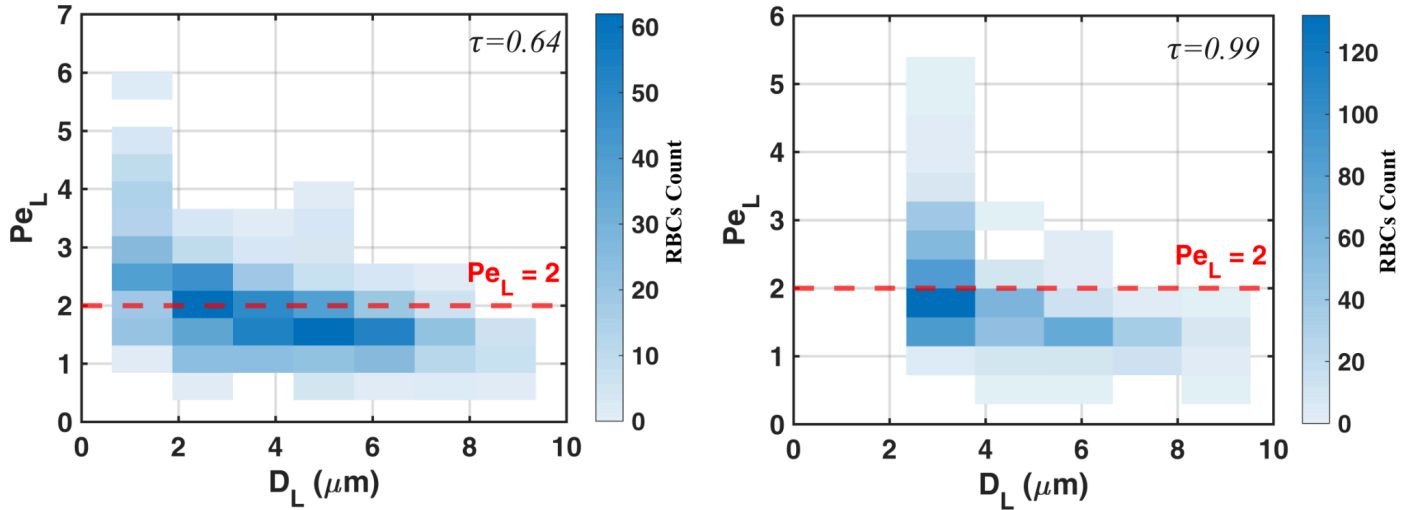


Figure 9. Binscatter plots of the RBC lingering Péclet number (Pe_L) versus the minimal distance from the bifurcation apex (D_L) for (a) discocytes and (b) spherocytes in the symmetric network.

- **Proportion of Lingering RBCs vs. Bifurcation Level.** In Figs. 10(a,c), we plot the fraction of RBCs with $Pe_L \geq 2$ (i.e., those classified as lingering cells) as a function of bifurcation level for the asymmetric and symmetric networks, respectively. Across all reduced areas τ , the proportion of lingering cells rises with each subsequent level, consistent with the cumulative positional-memory effect reported earlier (15, 25). Notably, even in the symmetric network (Fig. 10(c)), where geometric biases are minimized, the fraction of lingering cells still increases level by level. This underscores that apex interactions and progressive radial drift—rather than strictly geometry—are key drivers of lingering.
- **Average Lingering Intensity.** Figs. 10(b,d) display the average lingering $\langle Pe_L \rangle$ in each network as a function of bifurcation level. In both networks, $\langle Pe_L \rangle$ rises with level, confirming that RBCs not only linger more frequently but also remain for longer durations in downstream bifurcations. This finding is aligned with previous experimental work highlighting how red blood cells accumulate positional biases over multiple branching events (1, 3, 25, 31).
- **Comparison Between Asymmetric and Symmetric Networks.** Although the baseline lingering is somewhat higher in the symmetric network (Figs. 10(b,d))—likely due to the greater number of bifurcation levels—the overall trend of increasing Pe_L and rising proportion of lingerers is qualitatively similar in both networks. This corroborates the idea that local geometry is not the sole factor. Instead, RBC positional history and repeated interactions with the apex appear to be the dominant mechanism driving sustained lingering effects.

In summary, these results reinforce that RBCs with lower reduced areas linger more often and more intensely, but *all* RBCs, including near-spherical ones, exhibit increasing lingering over successive bifurcations. The history-driven accumulation of slight apex interactions emerges as a robust factor influencing microcirculatory blood flow, even in anatomically idealized (symmetric) networks.

FIGURE 10

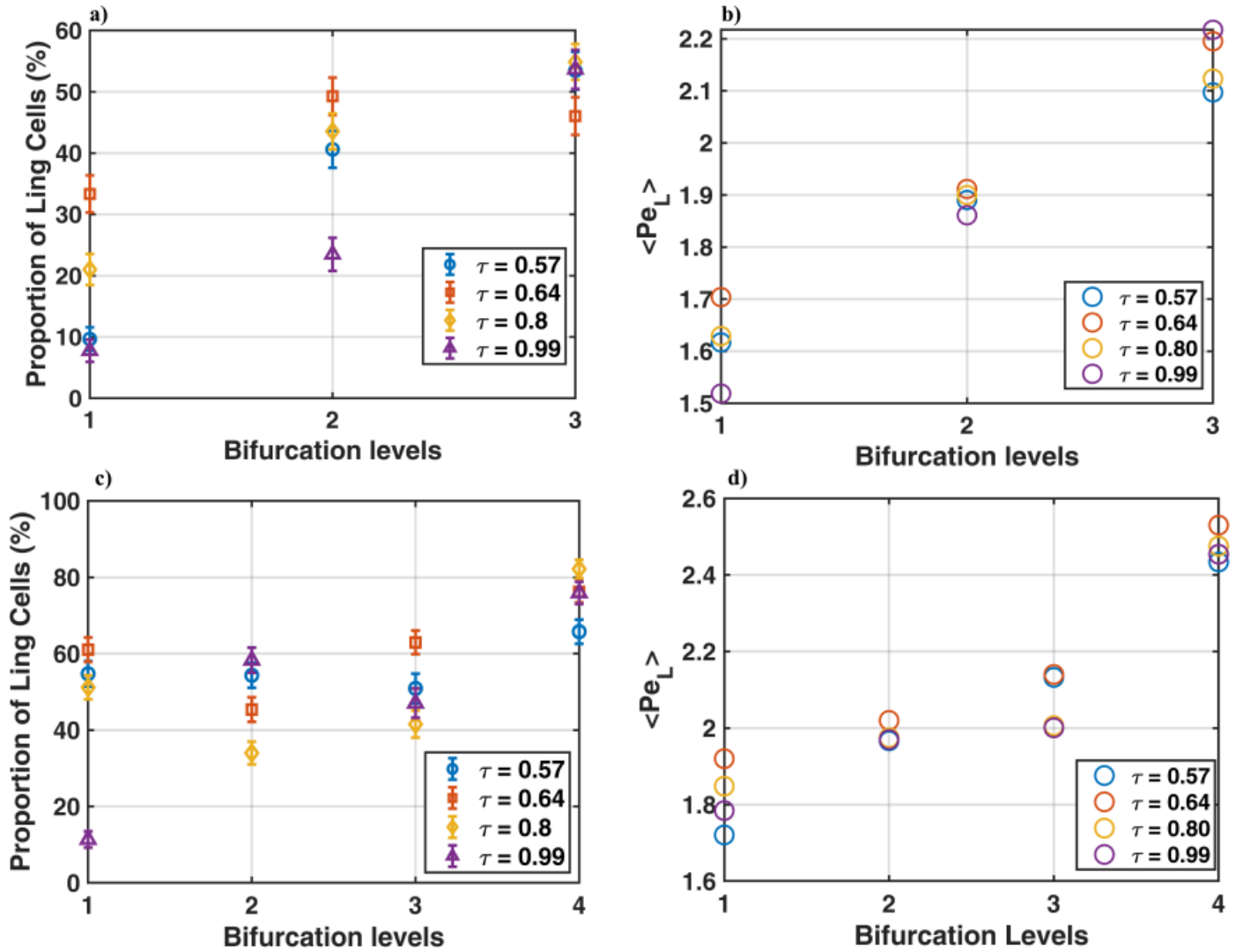


Figure 10. Proportion and intensity of lingering RBCs across bifurcation levels. Panels (a) and (c) present the proportion of RBCs classified as lingerers ($Pe_L \geq 2$) at each bifurcation level in the asymmetric and symmetric networks, respectively. Panels (b) and (d) depict the mean lingering $\langle Pe_L \rangle$ across bifurcation levels for the asymmetric and symmetric networks, respectively.

Lingering Behavior of Spherical RBCs

Although one might expect spherical RBCs (e.g., $\tau = 0.99$) to traverse bifurcations without lingering, our simulations reveal that these cells can still experience prolonged residence times in the apex region. Two main factors explain this phenomenon:

- **Mild deformation due to membrane flexibility.** Even “spherical” RBCs in our model retain a certain degree of flexibility. Figs. 11(e)–(h) illustrate that, despite having an equilibrium (relaxation) diameter of about $6 \mu\text{m}$, the RBCs’ *minimum* diameter can decrease to $\sim 5.5 \mu\text{m}$ and their *maximum* diameter can rise to $\sim 6.5 \mu\text{m}$ by the fourth bifurcation level. “Mild deformation” here refers to this measurable but sub-micron variation in cell diameter ($\lesssim 8\%$) induced by flow stresses near the apex; it is sufficient to cause non-negligible membrane–apex contact yet small compared to the gross shape changes seen for discocytes. This slight but non-negligible deformation implies that the RBC membrane still interacts with the bifurcation apex, imparting sufficient friction or tethering to increase the cells’ local residence time.
- **Apex geometry and multi-cell blockage.** In addition to membrane flexibility, the local geometry of the bifurcation apex (the “lingering triangle” shown in Figs. 11(b) and 11(f)) shrinks significantly at higher levels. As seen in Table 3, the triangle side lengths (e.g., “Side 2” and “Side 3”) decrease from $11 \mu\text{m}$ at Level 1 to $7.5 \mu\text{m}$ at Level 4, reducing the available cross-section through which RBCs can pass. Even when considering a spherical RBC with a maximum diameter

of $\sim 6.5 \mu\text{m}$, the proximity in size can result in transient blockages. “Crowding/blockage” denotes this multi-cell scenario: when two or more cells attempt to cross the apex simultaneously, their combined transverse footprint can exceed the local vessel diameter, transiently halting flow and prolonging the residence of upstream RBCs. Moreover, when *two* such cells attempt to cross simultaneously (Fig. 11(f) snapshot), their combined “footprint” can exceed the vessel diameter in the bifurcation, effectively halting flow and prolonging the residence of any RBCs upstream (see videos in supplementary materials). Consequently, local RBC concentration increases and self-organized blockage events become more probable, leading to higher lingering times.

As a comparison with Elongated RBCs ($\tau = 0.64$), For elongated RBCs (Figs. 11(a)–(d)), the minimal diameter can drop to $3\text{--}4 \mu\text{m}$, providing an alternative reason for lingering: cell edges can temporarily undergo physical hooking onto the apex (as defined above), but overall geometric blockage is less likely because their narrow dimension slips around the bifurcation more easily. In contrast, spherical RBCs rely less on physical hooking and more on crowding/blockage events, though both cell types display some degree of apex interaction and resulting residence.

Overall, these results show that apex lingering is not limited to elongated RBCs; spherical cells also exhibit measurable lingering when sufficiently flexible and when apex dimensions are restrictive. The interplay between slight shape deformation and tight apex geometry can slow or block local flow, meaning that a higher RBC positional memory and apex interactions remain relevant across a range of reduced areas τ .

Table 3: Bifurcation triangle areas and side lengths at each level.

	Level 1	Level 2	Level 3	Level 4
Side 1 (μm)	8.4	8.4	8.4	8.4
Side 2 (μm)	11	9.3	10	7.5
Side 3 (μm)	11	9.3	10	7.5
Triangle Area (μm^2)	42.69977	34.84988	38.11601	26.09752

FIGURE 11

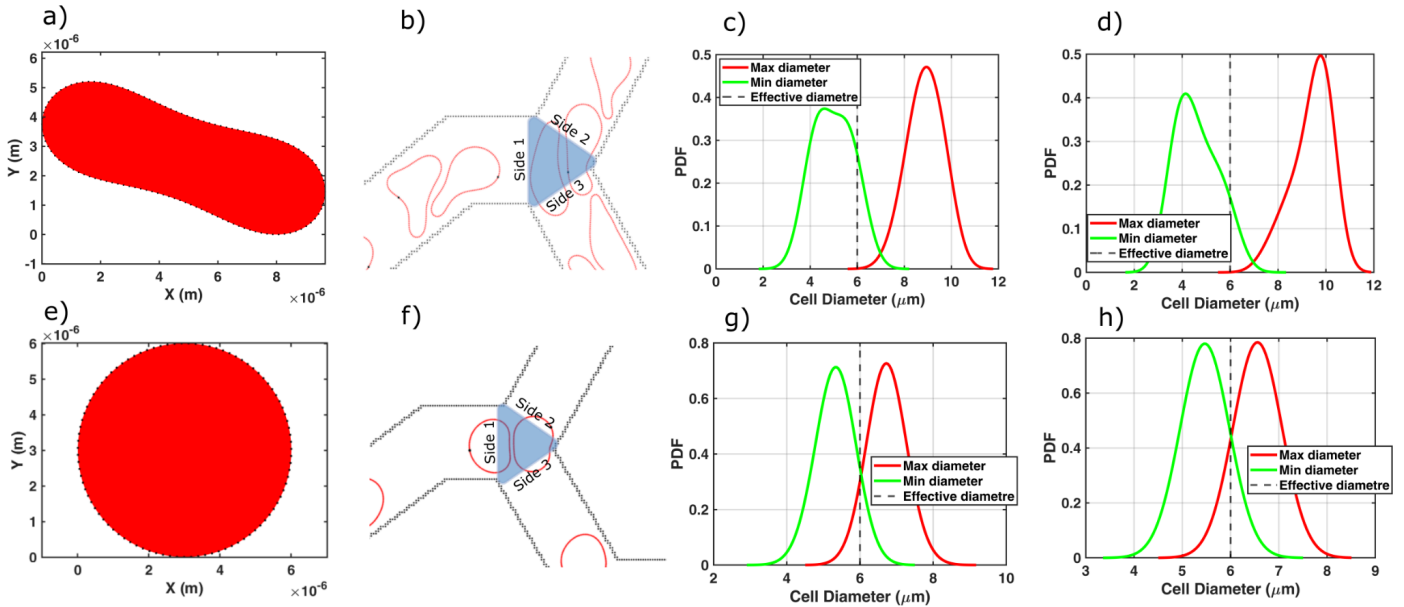


Figure 11. Lingering mechanisms for elongated and spherical RBCs. (a) Relaxation shape (discocyte): RBC with $\tau = 0.64$. (b) The “lingering triangle” region, where the first side equals the vessel diameter, and a snapshot of two lingering RBCs with $\tau = 0.64$. (c) Probability density function (PDF) of the maximum and minimum cell diameters in the bifurcation region for $\tau = 0.64$ at the *first* bifurcation level. (d) The same PDF for $\tau = 0.64$ at the *fourth* bifurcation level. (e) Relaxation shape (spherocyte): RBC with $\tau = 0.99$. (f) The lingering triangle region and a snapshot of two RBCs with $\tau = 0.99$. (g) PDF of the maximum and minimum cell diameters in the bifurcation region for $\tau = 0.99$ at the *first* bifurcation level. (h) The same PDF for $\tau = 0.99$ at the *fourth* bifurcation level.

CONCLUSION

In summary, we have numerically investigated how red blood cells distribute and linger across multiple bifurcations in microvascular networks, focusing on both asymmetric geometries (representative of real mesenteric vasculature) and artificially constructed symmetric networks. By combining a Lattice Boltzmann fluid solver with an immersed boundary approach for RBC membranes, we demonstrated several interconnected findings. Primarily, our results show that RBCs exhibit a cumulative positional-memory effect as they traverse successive bifurcation levels. By “positional memory” we mean that the transverse position and deformation state of a cell arriving at a bifurcation are set by its encounters with upstream bifurcation apices. Even when geometric asymmetry or vessel tapering is minimized, this effect causes lingering and deviations from the Zweifach-Fung hypothesis to grow significantly with each additional bifurcation. This highlights the primacy of RBC history—including repeated apex interactions, radial drifts, and collective effects—in driving non-classical microcirculatory phenomena.

Building on this concept, our investigation into cell shape revealed that while elongated (lower reduced area) RBCs tend to linger more strongly, even spherical cells ($\tau \approx 0.99$) experience substantial residence times, particularly when encountering tight apex geometries or multi-cell blockage events. This indicates that membrane flexibility and local crowding, not just cell elongation, are critical to lingering behavior. A direct consequence of this history effect is its impact on cell partitioning. We observed that deviations from the classical Zweifach-Fung hypothesis and the empirical law increase substantially over multiple bifurcations, suggesting that RBC positional history can overwhelm simple flow-based models at deeper network levels. Importantly, we also observed an increasing fraction of reversed-partitioning events at successive bifurcation levels, driven by the growing positional asymmetry of the incoming RBC column. This demonstrates that the history effect does not merely amplify classical partitioning but can qualitatively alter the partitioning regime. Our results therefore highlight that both local geometry and RBC shape must be accounted for to accurately predict RBC flux fractions in highly branched or pathological microvascular networks.

DATA AND CODE AVAILABILITY

The simulation data supporting the findings of this study and the code used to generate and analyze the results are available from the corresponding author upon reasonable request.

AUTHOR CONTRIBUTIONS

M.B., C.M., A.D., and C.W. conceived and designed the research. M.B. carried out all the simulations and performed the numerical analyses. M.B., C.M., A.D., and C.W. wrote the manuscript. M.B., C.M., A.D., Y.R., and C.W. discussed the results and critically revised the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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Red Blood Cell Positional History, Sphericity, and Network Architecture Influence
Lingering and Partitioning

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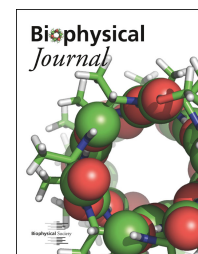
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